

Question 1

Which of the following statements regarding green hydrogen production from water electrolysis is correct?

- (a) Higher electrolyser efficiency reduces electricity consumption per unit of hydrogen produced
- (b) Electrolyser efficiency has no effect on hydrogen production costs
- (c) Hydrogen storage is usually the dominant contributor to levelised costs of hydrogen
- (d) Electrolysis converts hydrogen directly into electricity

Solution: (a)

Question 2

In most green hydrogen production systems, which component contributes the largest share of hydrogen costs?

- (a) Water cost
- (b) Electricity cost
- (c) Electrolysis CAPEX
- (d) Land lease cost

Solution: (b)

Question 3

Why can e-fuel production help reduce renewable energy curtailment in high-renewable power systems?

- (a) E-fuel plants can convert excess renewable electricity into storable chemical energy

- (b) E-fuels eliminate the need for electrolysis
- (c) Renewable generators stop producing electricity during low demand periods
- (d) E-fuels consume fossil fuels during production

Solution: (a)

Question 4

A renewable-energy-based hydrogen production system supplies green hydrogen for the production of three electrofuels: e-ammonia, e-methanol, and e-Fischer-Tropsch liquids (e-FTL). Using the techno-economic data provided below, calculate the levelised cost of hydrogen and subsequently determine the levelised costs of the three electrofuels. Finally, analyse the contribution of the different cost components to the final fuel costs.

Table 1: E-fuel plant techno-economic parameters

Fuel	hydrogen	e-methanol	e-ammonia	e-FTL
Plant capacity	10 MW _{el}	5 MW _{fuel}	4 MW _{fuel}	3 MW _{fuel}
Plant CAPEX (\$/kW)	1000	1200	900	1800
Fixed O&M cost (% CAPEX)	4%	4%	4%	4%
Plant efficiency (%)	70	—	—	—
Plant lifetime (Years)	30	30	30	30
Discount rate (%)	7	7	7	7
Annual operating hours	8000	6000	6000	6000
Power req. (MWh/MWh _{fuel})	—	0.03	0.14	0.05
CO ₂ req. (tCO ₂ /MWh _{fuel})	—	0.25	—	0.26
H ₂ consumption (MWh/MWh _{fuel})	—	1.2	1.1	1.5
Electricity price (\$/MWh)	40	40	40	40
CO ₂ price (\$/tCO ₂)	—	120	—	120

Solution:

For electrolysis plant, we would determine the total system CAPEX first before annualized CAPEX and fixed O&M costs with a capital recovery factor.

Therefore, the total system CAPEX = $(10 \times 1000 \times 1000)\$ = 10,000,000 \$$

$$\begin{aligned}
\text{Capital recovery factor} &= \frac{r(1+r)^n}{(1+r)^n - 1} \\
\text{Capital recovery factor} &= \frac{0.07.(1+0.07)^{30}}{(1+0.07)^{30} - 1} \\
&= 0.0806
\end{aligned}$$

Now, we will use this capital recovery factor for annualized CAPEX and fixed O&M cost:

Annualized CAPEX = total system CAPEX $\times$ Capital recovery factor

Annualized CAPEX = 10,000,000 \$ $\times$ 0.0806

Annualized CAPEX = 806,000 \$

Annualized fixed O&M = 806,000 \$ $\times$ 0.04

Annualized fixed O&M = 32,240 \$

Next, we will quantify annual electricity consumption and the associated costs:

Annual electricity consumption = 10 MW $\times$ 8000 h

Annual electricity costs = 10 MW $\times$ 8000 h $\times$ 40 \$/MWh

Annual electricity costs = 3,200,000 \$

Finally, we need the annual hydrogen production based on the efficiency of the plant:

Annual hydrogen production = 10 MW $\times$ 8000 h $\times$ 0.70

Annual hydrogen production = 56,000 MWh

The levelized costs of hydrogen = (Total annual system costs)/(Total hydrogen production)

The levelized costs of hydrogen = (3,200,000 + 806,000 + 32,240 \$) / (56,000 MWh)

The levelized costs of hydrogen = 72.11 \$/MWh

From the electrolysis plant, hydrogen production would cost 72.11 \$/MWh and the produced hydrogen acts as a feedstock for e-methanol and e-FTL.

In case of e-methanol,

Annualized CAPEX = 5 $\times$ 1200 $\times$ 1000 $\times$ 0.0806 \$

$$\text{Annualized CAPEX} = 483,600 \$$$

$$\text{Annualized O\&M} = 483,600 \times 0.04 \$$$

$$\text{Annualized O\&M} = 19,344 \$$$

$$\text{For annual electricity costs} = 5 \times 6000 \times 0.03 \times 40 = 36,000 \$$$

$$\text{For annual } CO_2 \text{ costs} = 5 \times 6000 \times 0.25 \times 120 = 900,000 \$$$

$$\text{For annual } H_2 \text{ costs} = 5 \times 6000 \times 1.2 \times 72.11 = 2,595,960 \$$$

$$\text{Levelized costs of e-methanol} = (483,600 + 19,344 + 36,000 + 900,000 + 2,595,960 \$) / (5 \times 6000 \text{ MWh})$$

$$\text{Levelized costs of e-methanol} = 134.49 \$/\text{MWh}$$

So, the levelized cost of methanol is found to be 134.49 \$/MWh. A similar calculation is to be performed for e-FTL plant, the only difference is the unit consumption of power and feedstocks.

$$\text{For e-FTL, Annualized CAPEX} = 3 \times 1800 \times 1000 \times 0.0806 \$$$

$$\text{Annualized CAPEX} = 435,240 \$$$

$$\text{Annualized O\&M} = 435,240 \times 0.04 \$$$

$$\text{Annualized O\&M} = 17,409 \$$$

$$\text{For annual electricity costs} = 3 \times 6000 \times 0.05 \times 40 = 36,000 \$$$

$$\text{For annual } CO_2 \text{ costs} = 3 \times 6000 \times 0.26 \times 120 = 561,600 \$$$

$$\text{For annual } H_2 \text{ costs} = 3 \times 6000 \times 1.5 \times 72.11 = 1,946,970 \$$$

$$\text{Levelized costs of e-FTL} = (435,240 + 17,409 + 36,000 + 561,600 + 1,946,970 \$) / (3 \times 6000 \text{ MWh})$$

$$\text{Levelized costs of e-FTL} = 166.51 \$/\text{MWh}$$

Due to the higher requirement of power, CO_2 , H_2 , e-FTL begs the highest costs among the e-fuels of our focus.

$$\text{Lastly, for e-ammonia, annualized CAPEX} = 4 \times 900 \times 1000 \times 0.0806 \$$$

$$\text{Annualized CAPEX} = 290,160 \$$$

$$\text{Annualized O\&M} = 290,160 \times 0.04 \$$$

Annualized O&M = 11,606 \$

For annual electricity costs = $4 \times 6000 \times 0.14 \times 40 = 134,400$ \$

For annual H_2 costs = $4 \times 6000 \times 1.1 \times 72.11 = 1,903,704$ \$

Levelized costs of e-ammonia = $(290,160 + 11,606 + 134,400 + 1,903,704 \text{ $}) / (4 \times 6000 \text{ MWh})$

Levelized costs of e-ammonia = 97.49 \$/MWh